Reconstructing the Goldberg–Palatini Action from Noether Identities

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(Dated: October 17, 2026)

Abstract

We investigate the reconstruction of the first-order gravitational action from the Noether identities associated with diffeomorphism invariance. The independent fields are a symmetric perturbation $\phi^{\mu\nu}$ of the contravariant metric density and a trace-adjusted connection $G^{\lambda}{}_{\mu\nu}$, with prescribed transformation laws. In generic spacetime dimension, we construct a local, Lorentz-covariant ansatz containing quadratic terms with at most two derivatives and a cubic interaction restricted to the structure ϕGG, whose kernels are expanded in tensor bases comprising 31 coefficients. The Noether identities, expanded order by order in the fields, yield a linear system that eliminates the $\phi\phi$ kernel and determines the remaining coefficients up to an overall normalization, recovering the Goldberg–Palatini action. Since the connection can be eliminated exactly, this also fixes the complete second-order action, with its infinite set of graviton vertices. The construction requires no choice of energy–momentum tensor, the source of ambiguity in the self-coupling approach, and uniqueness holds within the assumed class of actions. The kernel bases, the identities and the solution are given in appendices, and the symbolic computations are publicly available.

I. INTRODUCTION

The extent to which gravitational dynamics can be recovered from the consistency of an interacting massless spin-two field is a longstanding question. In the self-coupling construction, which goes back to Feynman [1] and was completed by Deser [2], the use of first-order variables [3, 4] allows the nonlinear completion to be expressed through a cubic interaction, avoiding the infinite expansion of the metric formulation. Discussions of this approach have also emphasized the assumptions entering the construction, the ambiguities in the energy–momentum tensor, and the treatment of boundary terms [5, 6]. These issues motivate a formulation in which the constraints imposed by gauge symmetry can be examined directly.

Local gauge invariance implies differential identities among the Euler–Lagrange derivatives of the action. These are identities of Noether's second theorem and hold off shell. Once the field transformations and a class of admissible actions have been specified, they can be used as algebraic conditions on unknown interaction coefficients. This reverses the usual

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direction of the calculation: instead of deriving identities from a known action, one uses the identities to determine which actions are compatible with the prescribed transformations. This strategy was used in Ref. [7] to determine the scalar–graviton vertex in noncommutative space, where the closed form of the action is not available, from the invariance under a restricted class of coordinate transformations.

In this work we apply this strategy to the first-order gravitational variables $\phi^{\mu\nu}$ and $G^{\lambda}{}_{\mu\nu}$, the perturbation of the contravariant metric density and a trace-adjusted connection treated as an independent field. The same strategy applies in the metric formulation, where the field-dependent part of the transformation is essential: invariance under the linearized transformations alone fixes only part of each vertex, whereas the full transformation determines the three-graviton vertex from the quadratic term, up to total derivatives [8]. The program does not terminate there, however, since new vertices appear at every order. The first-order choice is adapted to the polynomial structure of the Goldberg–Palatini action [3, 4, 9] and to its use in perturbative gravity [10]. We restrict the action to quadratic kernels of prescribed derivative order and to a derivative-free cubic interaction of the form ϕGG, linear in ϕ and quadratic in G as in the Goldberg–Palatini action, expand the kernels in tensor bases, and impose the Noether identities order by order in the fields; the determination of the action then reduces to a finite system of linear constraints on 31 coefficients, which is solved completely. Because the field equation of G is algebraic, the connection can then be eliminated exactly, so the finite determination of the first-order action also fixes the infinite set of graviton vertices of the metric formulation [10].

Our objective is to determine the freedom remaining in this ansatz after the Noether identities are imposed. Locality, Lorentz covariance and parity, the transformation laws, the field content, the coupling-power assignment that bounds the number of derivatives, and the restriction on the cubic interaction are inputs to the construction; the uniqueness statement concerns this specified class of actions and does not assert that diffeomorphism invariance alone selects Einstein gravity among all local gravitational theories.

The construction complements the self-coupling approach of Refs. [2, 6]. Self-coupling combines the conserved current of global translations with the consistency of the gauge identities of the spin-two field, and thus requires the choice of an energy–momentum tensor, the point at issue in Ref. [5]. Its recent extension to metric-affine theories [11] identifies this choice as the source of the ambiguities, notes that in the first-order construction of

Ref. [2] the connection is an auxiliary field that is not itself bootstrapped, and observes that the existing arguments establish existence rather than uniqueness. In the present work no energy–momentum tensor is constructed: the off-shell identities of Noether's second theorem are imposed directly, the geometrical input being retained in the transformation laws; the variations of both independent fields enter the identities; and uniqueness is established within the stated class. The idea of using the consistency of the gauge identities instead of self-coupling goes back to Wald [12], who analyzed the consistent nonlinear extensions through identities among the field equations; here the nonlinear transformations are prescribed and the compatible action is determined within a finite ansatz. The result concerns the bulk action up to total derivatives and does not determine boundary terms.

The paper is organized so that the main text carries the logic of the construction and the appendices carry its details. Section II fixes the fields and their transformations, Section IV the ansatz and its counting, Section V the identities and their decomposition into sectors, Section VI the solution, and Section VII the reduction to the metric formulation. Figure 1 summarizes the procedure. The notation and the derivation of the transformation laws are collected in Appendix A; explicit tensors, bases, identities and kernels in Appendices B–E; and Appendix F describes the symbolic implementation and how to reproduce every check.

II. FIRST-ORDER VARIABLES AND GAUGE TRANSFORMATIONS

a. Conventions.

- Units are $\hbar = c = 1$, so that dimensions are mass dimensions; the objects used throughout and their dimensions are listed in Table V of Appendix A.

- Spacetime has d dimensions and flat metric $\eta_{\mu\nu}$ of signature $(+,-,\ldots,-)$. The algebraic tensor identities used below are polynomial in η and the momenta and do not depend on the signature; for generic d, $\sqrt{-g}$ is read as $\sqrt{|g|}$.

- Index symmetrization is not normalized: $h^{(\mu\nu)} \equiv h^{\mu\nu} + h^{\nu\mu}$.

- Fourier transforms are $\tilde{\phi}^{\mu\nu}(p) = \int d^dx\, e^{ip\cdot x}\phi^{\mu\nu}(x)$, and likewise for $\tilde{G}^{\lambda}{}_{\mu\nu}$ and for the gauge parameter $\tilde{\omega}^{\mu}$, with $\int_p \equiv \int d^dp/(2\pi)^d$. A derivative acting on a field of momentum p becomes $-ip_\mu$, and a product of fields becomes a convolution.

- The gravitational coupling κ satisfies $\kappa^2 \propto G_N$, so that $[\kappa] = (2-d)/2$; its normalization is fixed in Section VI, where the solution is identified with $\kappa^{-2}\int d^dx\, h^{\mu\nu}R_{\mu\nu}(\Gamma)$.

b. Compact notation. In the main text indices are suppressed and a field carries its momentum as a subscript: ϕ_p, G_p and ω_p stand for $\tilde\phi^{\mu\nu}(p)$, $\tilde G^\lambda{}_{\mu\nu}(p)$ and $\tilde\omega^\mu(p)$. A kernel $K(A,B)$ or $K(A,B,C)$ denotes the full contraction of each argument with the corresponding index group of K, for instance $K^{\phi G}(\phi_{-p},G_p) \equiv \tilde\phi^{\mu\nu}(-p)\,K^{\phi G}_{\mu\nu\lambda}{}^{\rho\sigma}(p)\,\tilde G^\lambda{}_{\rho\sigma}(p)$. The field-independent and field-linear parts of the transformations are written as $D_{\phi^0}(p)\,\omega_p$, linear in ω, and $D_{\phi^1}(p,q)(\phi_{p-q},\omega_q)$, bilinear in the field and ω, and likewise for G; the label names the field produced and the order in κ (Eq. (2.5) below and Appendix B).

c. Fields. The independent fields are the perturbation of the Goldberg density [9] and the trace-adjusted connection, both normalized with the gravitational coupling κ,

$$\kappa\,\phi^{\mu\nu} = \sqrt{-g}\,g^{\mu\nu} - \eta^{\mu\nu}, \qquad \kappa\,G^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \tfrac{1}{2}\big(\delta^\lambda_\mu\Gamma^\sigma{}_{\nu\sigma} + \delta^\lambda_\nu\Gamma^\sigma{}_{\mu\sigma}\big), \tag{2.1}$$

both real and symmetric in $\mu\nu$. Here $\Gamma^\lambda{}_{\mu\nu}$ is an independent symmetric affine connection, the Palatini variable [3, 4], and not the Levi-Civita connection built from $g_{\mu\nu}$: for the action found below, the field equation of Γ identifies the two when $d>2$ and the metric is nondegenerate, and that equation is not imposed in the construction. The second relation in Eq. (2.1) is a linear change of variables, invertible for $d\neq 1$. Taking its trace gives $\Gamma^\sigma{}_{\nu\sigma} = \frac{2\kappa}{1-d}\,g_\nu$, with $g_\nu \equiv G^\sigma{}_{\nu\sigma}$, and substituting back,

$$\Gamma^\lambda{}_{\mu\nu} = \kappa\Big[G^\lambda{}_{\mu\nu} + \frac{1}{1-d}\big(\delta^\lambda_\mu\,g_\nu + \delta^\lambda_\nu\,g_\mu\big)\Big], \tag{2.2}$$

Eqs. (6)–(8) of Ref. [13]. The trace adjustment is what cancels the $\partial_\nu\Gamma^\lambda{}_{\mu\lambda}$ term of $h^{\mu\nu}R_{\mu\nu}(\Gamma)$ against the trace part of $\partial_\lambda\Gamma^\lambda{}_{\mu\nu}$, leaving the single derivative term of Eq. (6.1) below. Like ϕ, which is the metric-density perturbation divided by κ and has mass dimension $(d-2)/2$, G is the trace-adjusted connection divided by κ and has mass dimension $d/2$, instead of the dimension 1 of a connection. The definitions in Eqs. (2.1)–(2.2) serve only to fix the transformation laws below; from then on ϕ and G are treated as independent fields, and no relation between G and a metric is assumed.

d. Transformations. Under an infinitesimal diffeomorphism $x^\mu \to x^\mu - \kappa\,\omega^\mu(x)$ the fields transform as $\delta\phi = \delta^{(0)}\phi + \kappa\,\delta^{(1)}\phi$ and $\delta G = \delta^{(0)}G + \kappa\,\delta^{(1)}G$, with

$$\delta^{(0)}\phi^{\mu\nu} = \eta^{\mu\nu}\partial\cdot\omega - \partial^{(\mu}\omega^{\nu)}, \qquad \delta^{(1)}\phi^{\mu\nu} = \partial_\rho(\phi^{\mu\nu}\omega^\rho) - \partial_\rho\omega^{(\mu}\phi^{\nu)\rho}, \tag{2.3}$$

$$\delta^{(0)}G^\lambda{}_{\mu\nu} = \partial_\mu\partial_\nu\omega^\lambda - \tfrac{1}{2}\delta^\lambda_{(\mu}\partial_{\nu)}\partial\cdot\omega, \quad \delta^{(1)}G^\lambda{}_{\mu\nu} = \omega^\rho\partial_\rho G^\lambda{}_{\mu\nu} - G^\rho{}_{\mu\nu}\partial_\rho\omega^\lambda + G^\lambda{}_{\rho(\mu}\partial_{\nu)}\omega^\rho. \tag{2.4}$$

The split into a field-independent contribution $\delta^{(0)}$ and a field-linear contribution $\kappa\delta^{(1)}$ is what organizes the whole construction. Eq. (2.3) follows from the transformation of $h^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$ as a tensor density of weight one, with the background $\eta^{\mu\nu}$ held fixed; Eq. (2.4) follows from the transformation of an affine connection through Eq. (2.1). Both are derived in Appendix A; they are Eqs. (17b) and (18) of Ref. [13]. These two laws are the only input of the construction beyond the class of actions defined in the next section.

In momentum space the derivatives become momenta and $\delta^{(1)}$ becomes a convolution:

$$\delta\phi_p = D_{\phi^0}(p)\,\omega_p + \kappa\int_q D_{\phi^1}(p,q)\,(\phi_{p-q},\omega_q), \qquad \delta G_p = D_{G^0}(p)\,\omega_p + \kappa\int_q D_{G^1}(p,q)\,(G_{p-q},\omega_q). \tag{2.5}$$

The four tensors $D_{\phi^0}, D_{\phi^1}, D_{G^0}, D_{G^1}$ are listed in Appendix B.

III. POLYNOMIAL CLOSURE

Before the kernels are specified, the transformation laws already restrict which field structures a polynomial action can contain. The argument uses Eqs. (2.3)–(2.4) only for a particular choice of ω^μ.

a. Coupling assignment and derivative counting. Since $\kappa\omega$ is a coordinate displacement, $[\kappa\omega] = -1$, and Eqs. (2.3)–(2.4) give $[\phi] = (d-2)/2$ and $[G] = d/2$, with $[\kappa] = (2-d)/2$. We consider local actions

$$S = \sum_{n=2}^{N} \kappa^{n-2} S_n, \tag{3.1}$$

where S_n is homogeneous of degree n in the fields, Lorentz covariant and parity even, built with η and derivatives, and contains no dimensionful constant; N is finite and is the highest degree for which $S_N \neq 0$ as a functional, that is, up to total derivatives. The power κ^{n-2} and the absence of any other dimensionful constant are both needed: additional powers of κ at fixed n, or a new dimensionful parameter, would allow terms with more derivatives and the correct dimension.[1] A local monomial $\mathcal{L}_{a,b,r}$ with a fields ϕ, b fields G and r derivatives, multiplied by κ^{a+b-2}, has mass dimension $d+b+r-2$, so that a dimensionless action requires

$$b + r = 2. \tag{3.2}$$

[1] Equivalently, the action is κ^{-2} times a functional of $h - \eta = \kappa\phi$ and κG that is independent of κ and contains no other dimensionful constant; a term of degree n in ϕ, G then carries κ^{n-2}.

Thus $\phi\phi$ carries two derivatives, ϕG one and GG none; at cubic order $\phi\phi\phi$ carries two, $\phi\phi G$ one and ϕGG none, and GGG is excluded. The counting does not limit the number of fields ϕ.

b. The highest-order identity. The transformation $\delta = \delta^{(0)} + \kappa\delta^{(1)}$ is exactly affine in the fields: $\delta^{(0)}$ lowers the degree by one and $\delta^{(1)}$ preserves it. Collecting powers of κ in $\delta S = 0$ gives

$$\delta^{(0)} S_2 = 0, \qquad \delta^{(0)} S_{n+1} + \delta^{(1)} S_n = 0 \quad (2 \le n \le N-1), \qquad \delta^{(1)} S_N = 0. \tag{3.3}$$

The last identity has no compensating term, because there is no S_{N+1}. These are identities between integrated functionals, that is, between local densities up to total derivatives, and every statement below that a sector vanishes is meant in this sense.

c. A dilatation. The identities hold for every ω^μ, in particular for $\omega^\mu = \varepsilon x^\mu$, which for $|\kappa\varepsilon| \ll 1$ is the dilatation $x^\mu \to (1 - \kappa\varepsilon)x^\mu$.[2] This is a particular diffeomorphism, not an additional symmetry: it gives necessary conditions, which suffice to exclude sectors but not to construct the action. With $\partial_\nu \omega^\mu = \varepsilon \delta^\mu_\nu$ and $\partial_\alpha \partial_\beta \omega^\mu = 0$, Eqs. (2.3)–(2.4) give

$$\begin{aligned} \delta^{(1)}\phi &= \varepsilon\,(x\cdot\partial + d - 2)\,\phi, & \delta^{(1)} G &= \varepsilon\,(x\cdot\partial + 1)\,G, \\ \delta^{(0)}\phi^{\mu\nu} &= \varepsilon\,(d-2)\,\eta^{\mu\nu}, & \delta^{(0)} G &= 0; \end{aligned} \tag{3.4}$$

$\delta^{(0)} G$ vanishes because the inhomogeneous part of the transformation of a connection involves second derivatives of ω. The numbers $d-2$ and 1 are weights under this transformation, not mass dimensions. Each derivative adds one unit to the weight, since ∂_α acting on x^ρ gives δ^ρ_α, and contractions with the fixed background η add nothing. For a local density $\mathcal{L}_{a,b,r}$ therefore $\delta^{(1)}\mathcal{L}_{a,b,r} = \varepsilon\,[\,x\cdot\partial + a(d-2) + b + r\,]\,\mathcal{L}_{a,b,r}$, and since $\int d^dx\, x\cdot\partial\mathcal{L} = -d\int d^dx\,\mathcal{L}$, the integrated functional $S_{a,b,r}$ satisfies

$$\delta^{(1)} S_{a,b,r} = \varepsilon\,[\,a(d-2) + b + r - d\,]\,S_{a,b,r} = \varepsilon\,(a-1)(d-2)\,S_{a,b,r}, \tag{3.5}$$

where Eq. (3.2) was used in the last step. Table I lists the values for the sectors with two and three fields.

[2] To avoid conditions on a parameter that grows at infinity, take smooth fields of compact support and $\omega^\mu = \varepsilon\chi(x)x^\mu$, with χ smooth, of compact support, and equal to one on a neighbourhood of the supports of the fields. All terms of the identities lie in that neighbourhood, where the derivatives of χ vanish, and integrations by parts leave no boundary terms.

Sector	$\phi\phi$	ϕG	GG	ϕGG	$\phi\phi G$	$\phi\phi\phi$
(a,b,r)	$(2,0,2)$	$(1,1,1)$	$(0,2,0)$	$(1,2,0)$	$(2,1,1)$	$(3,0,2)$
$\delta^{(1)}S_{a,b,r}/(\varepsilon S_{a,b,r})$	$d-2$	0	$-(d-2)$	0	$d-2$	$2(d-2)$

TABLE I. Weights of the sectors with two and three fields under the dilatation $\omega^\mu = \varepsilon x^\mu$, Eq. (3.5).

d. Termination at cubic order. In the highest-order identity, Eq. (3.5) gives

$$\delta^{(1)}S_N = \varepsilon\,(d-2)\sum_{a+b=N}(a-1)\,S_{a,b,2-b} = 0. \tag{3.6}$$

Sectors with different a have different field content, and the identity holds for arbitrary independent fields (rescaling $\phi \to u\phi$ and $G \to vG$ separates them), so for $d > 2$ every sector with $a \neq 1$ vanishes. The surviving sector has $a = 1$, hence $b = N-1$ and $r = 3-N$, and $r \geq 0$ requires $N \leq 3$. A polynomial action therefore terminates at most at cubic order, and its cubic part can only be the derivative-free vertex ϕGG; the sectors $\phi\phi G$, with a derivative on either field, and $\phi\phi\phi$ are excluded. The same mechanism forbids closure in the metric formulation: written in the density $h^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$ alone, every sector has $b = 0$ and $r = 2$, and the highest one, with $a = N \geq 2$, has nonvanishing weight, so its variation must be compensated at the next order and the series does not terminate [8]. The weights depend on the variables: for the covariant perturbation $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ the dilatation gives $h_{\mu\nu}$ weight 2, and Eq. (3.5) does not apply.

e. The identity of order κ. With $S_3 = S_{1,2,0}$, the identity $\delta^{(0)}S_3 + \delta^{(1)}S_2 = 0$ can be evaluated in the same way. The quadratic action $S_2 = S_{2,0,2} + S_{1,1,1} + S_{0,2,0}$ contains the sectors $\phi\phi$, ϕG and GG, with weights $d-2$, 0 and $-(d-2)$. Since S_3 is linear in ϕ and contains no derivatives, Eq. (3.4) gives $\delta^{(0)}S_3 = \varepsilon(d-2)\,S_3|_{\phi\to\eta}$, the functional quadratic in G obtained by replacing $\phi^{\mu\nu}$ with $\eta^{\mu\nu}$. The identity becomes $\varepsilon(d-2)\big[S_{2,0,2} - S_{0,2,0} + S_3|_{\phi\to\eta}\big] = 0$, and since the first functional depends only on ϕ and the others only on G, for $d > 2$,

$$S_{2,0,2} = 0, \qquad S_{0,2,0} = S_3\big|_{\phi\to\eta}. \tag{3.7}$$

The quadratic action has no $\phi\phi$ sector, and its algebraic GG term is the cubic vertex with ϕ replaced by η, as when both come from a single structure $h^{\mu\nu}(GG)_{\mu\nu}$ with $h^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$.

f. Result. *Let $d > 2$, and let S be a local action of the form (3.1), Lorentz covariant, parity even, polynomial in $\phi^{\mu\nu}$ and $G^\lambda{}_{\mu\nu}$, with no dimensionful constant other than κ, and*

invariant up to total derivatives under Eqs. (2.3)–(2.4). *Then* $N \leq 3$, *the cubic part contains only the derivative-free structure* ϕGG, *and the quadratic part satisfies Eq.* (3.7). The transformation laws are inputs; what is shown is that they restrict the action. The dilatation is blind to the tensor structure, since all contractions within a sector carry the same weight. It therefore decides which sectors may appear and relates whole sectors to one another. The ϕG sector has weight zero and is not constrained by the dilatation, and the ϕGG vertex is related only as a whole to the GG term by Eq. (3.7); the contractions in the ϕG, GG and ϕGG kernels and their relative normalization are fixed only by the Noether identities with general ω^μ, Sections V and VI.

IV. THE ACTION ANSATZ

By the result of Section III, the most general action in the class considered is $S = S_2 + \kappa S_3$ with

$$\begin{aligned} S_2 &= \int_p \Big[K^{\phi\phi}(\phi_{-p}, \phi_p) + K^{\phi G}(\phi_{-p}, G_p) + K^{GG}(G_{-p}, G_p) \Big], \\ S_3 &= \int_p \int_q K^{\phi GG}(\phi_{p+q}, G_{-p}, G_{-q}). \end{aligned} \tag{4.1}$$

By Eq. (3.2), $K^{\phi\phi}$ is quadratic and $K^{\phi G}$ linear in the momentum p, while K^{GG} and $K^{\phi GG}$ are momentum independent. The cubic vertex $K^{\phi GG}$ is purely algebraic, which is the feature of the first-order formulation that makes the determination finite. We keep the full quadratic part and do not impose Eq. (3.7): the Noether identities reproduce it (Section VI), which provides an independent check.

a. Tensor bases. Each kernel is expanded in a basis of tensors built from $\eta^{\mu\nu}$ and p^μ, with the index symmetries inherited from the fields it multiplies; parity-odd structures involving the Levi-Civita tensor are not considered. The bases are generated by symmetrizing, under the index symmetries of each kernel, the tensor monomials, that is, the products of η's and p's with the required indices and degree in p; their linear independence is verified through Gram determinants (Appendix C). Table II gives the outcome. For $d \geq 4$ the four bases are independent and contain $4 + 6 + 5 + 16 = 31$ unknown coefficients in total.

Kernel	Index symmetries	Momentum	Monomials	Basis dimension
$K^{\phi\phi}_{\mu\nu\rho\sigma}$	$\mu \leftrightarrow \nu$, $\rho \leftrightarrow \sigma$, $(\mu\nu) \leftrightarrow (\rho\sigma)$	p^2	$p^2\eta\eta$, $pp\eta$	4
$K^{\phi G}_{\mu\nu\lambda\rho\sigma}$	$\mu \leftrightarrow \nu$, $\rho \leftrightarrow \sigma$	p	$p\eta\eta$	6
$K^{GG}_{\lambda\mu\nu\rho\sigma\tau}$	$\mu \leftrightarrow \nu$, $\sigma \leftrightarrow \tau$, $(\lambda\mu\nu) \leftrightarrow (\rho\sigma\tau)$	p^0	$\eta\eta\eta$	5
$K^{\phi GG}_{\mu\nu\lambda\alpha\beta\rho\sigma\tau}$	$\mu \leftrightarrow \nu$, $\alpha \leftrightarrow \beta$, $\sigma \leftrightarrow \tau$, $(\lambda\alpha\beta) \leftrightarrow (\rho\sigma\tau)$	p^0	$\eta\eta\eta\eta$	16

TABLE II. The four kernels of the ansatz (4.1), their symmetries, and the dimension of the tensor basis in which each is expanded, for $d \geq 4$.

V. NOETHER IDENTITIES AND THEIR DECOMPOSITION

Invariance of S under Eqs. (2.3)–(2.4), ordered by powers of κ, or equivalently by the number of fields, gives three identities,

$$\delta^{(0)} S_2 = 0, \qquad \delta^{(0)} S_3 + \delta^{(1)} S_2 = 0, \qquad \delta^{(1)} S_3 = 0, \tag{5.1}$$

where $\delta^{(m)} S_n \equiv \int (\delta S_n / \delta A)\, \delta^{(m)} A$ summed over the fields $A = \phi, G$. Each identity holds for arbitrary ω and arbitrary field configurations, so it can be decomposed further according to which fields appear. Taking functional derivatives with respect to the fields and setting the fields to zero turns each piece into an algebraic tensor identity among the kernels, in momentum space. The decomposition is summarized in Table III; the six identities are written out in Appendix D. The resulting relations among the action kernels are also called classical Ward identities, the terminology used in the accompanying notebooks; we keep "Noether identities" to emphasize their derivation from Noether's second theorem. They concern the classical action before gauge fixing, not the Green functions or the BRST identities of a gauge-fixed theory.

The identities $\mathcal{W}^{\phi}$ and $\mathcal{W}^{G}$, which involve only $\delta^{(0)}$ and S_2, are labeled by the field sector; in the others, which involve $\delta^{(1)}$ or S_3, the superscript 3 follows the accompanying notebook, where these are grouped as cubic identities, and the second digit numbers the field sectors $\phi\phi$, ϕG, GG and ϕGG. Two remarks on the structure of Table III contain most of the argument. First, $\mathcal{W}^{31}$ involves the $\phi\phi$ kernel alone, because S_3 contains two factors of G and $\delta^{(1)} G$ is linear in G: nothing in the cubic sector can contribute to the $\phi\phi$ sector. This follows

Identity	From	Field sector	Kernels involved	Projection basis
$\mathcal{W}^{\phi}$	$\delta^{(0)}S_2$	ϕ	$K^{\phi\phi}$, $K^{\phi G}$	3
$\mathcal{W}^{G}$	$\delta^{(0)}S_2$	G	$K^{\phi G}$, K^{GG}	6
$\mathcal{W}^{31}$	$\delta^{(1)}S_2$	$\phi\phi$	$K^{\phi\phi}$	7
$\mathcal{W}^{32}$	$\delta^{(0)}S_3 + \delta^{(1)}S_2$	ϕG	$K^{\phi GG}$, $K^{\phi G}$	23
$\mathcal{W}^{33}$	$\delta^{(0)}S_3 + \delta^{(1)}S_2$	GG	$K^{\phi GG}$, K^{GG}	20
$\mathcal{W}^{34}$	$\delta^{(1)}S_3$	ϕGG	$K^{\phi GG}$	95

TABLE III. Decomposition of the Noether identities (5.1) into field sectors. The last column is the generic-dimension size of the projection basis onto which each identity is projected to extract scalar equations for the coefficients, with the momenta set equal as described in the text; dimension-specific dependencies, such as that of the 95-element basis at $d = 4$, are discussed in Appendix D.

from the restriction of S_3 to ϕGG: $\phi\phi\phi$ or $\phi\phi G$ terms would contribute to this sector through $\delta^{(0)}\phi$ and $\delta^{(0)}G$. Second, the cubic kernel $K^{\phi GG}$ enters $\mathcal{W}^{32}$ and $\mathcal{W}^{33}$, which relate it to the quadratic kernels through the field-dependent part $\delta^{(1)}$ of the transformations, and $\mathcal{W}^{34}$, which constrains the cubic kernel alone. Within the chosen ansatz, these identities couple the cubic vertex to the quadratic action; the solution below establishes that no independent cubic coefficient remains.

Each identity is turned into scalar equations by projecting it onto a basis of tensors with the same index structure and momentum dependence; since the identity vanishes, so does every projection. Conversely, for an identity that can be expanded in a projection basis with nonsingular Gram matrix, the vanishing of all projections is equivalent to the vanishing of the identity; the one exception, $\mathcal{W}^{34}$ at $d = 4$, is discussed in Appendix D.

a. Identified momenta. In the identities with two or three momenta the momenta are set equal to one another, $p_1 = p_2 = p_3 \equiv p$, with p itself arbitrary, so that every projection basis is built from η and a single vector; the dimensions in Table III refer to these bases. $\mathcal{W}^{\phi}$ and $\mathcal{W}^{G}$ involve a single momentum and are used as they stand. The Noether identities hold as tensor identities for arbitrary momenta, in particular for equal ones, so the restriction to equal momenta can only weaken the system: it yields necessary conditions, never spurious ones, and the solution space of the equal-momentum system contains that of the full one. The equal-momentum system already has a one-dimensional solution space (Section VI),

spanned by the Goldberg–Palatini action. For this action the identities hold for arbitrary momenta, since $h^{\mu\nu}R_{\mu\nu}(\Gamma)$ is a scalar density under Eqs. (2.3)–(2.4) by construction and its variation is a total derivative, which integrates to zero for gauge parameters of compact support. The two solution spaces therefore coincide, and nothing is lost by the identification.

VI. SOLUTION

The identities are generated and the resulting linear system is solved symbolically with FeynCalc [14] in Mathematica [15] (Appendix F). The system is solved in a sequence that follows the structure of Table III, shown in Table IV.

Step	Identity	Constraints imposed	Remaining freedom
1	$\mathcal{W}^{31}$	all 4 of $K^{\phi\phi}$, to zero	27
2	$\mathcal{W}^{\phi}$, $\mathcal{W}^{G}$	9 of the 11 in $K^{\phi G}$, K^{GG}	2 in S_2; 16 in $K^{\phi GG}$
3	$\mathcal{W}^{32}$	the remaining relative freedom in S_2; all 16 of $K^{\phi GG}$	1 (overall constant)
4	$\mathcal{W}^{33}$, $\mathcal{W}^{34}$	none: both are satisfied identically	1

TABLE IV. Sequential solution of the constraint system. The third column counts independent constraints, that is, parameters eliminated at each step; which individual coefficients are expressed in terms of which depends on the parametrization chosen.

Step 1 shows that, within the ansatz, the identities force the purely metric kernel to vanish, $K^{\phi\phi} = 0$, before any comparison with a known action: in this formalism the dynamics of ϕ is carried entirely by its coupling to G. At quadratic order alone, $\mathcal{W}^{\phi}$ and $\mathcal{W}^{G}$ would allow a $K^{\phi\phi}$ of Fierz–Pauli form, invariant by itself under $\delta^{(0)}\phi$; it would merely duplicate the graviton kinetic term generated by the elimination of G, Eq. (7.1), and it is the field-dependent identity $\mathcal{W}^{31}$ that removes it. Steps 2 and 3 then leave only the overall constant, the cubic identity in the ϕG sector fixing both the relative coefficient left in S_2 and all sixteen coefficients of $K^{\phi GG}$, and Step 4 imposes no further condition. The solution is therefore unique up to an overall constant.

Transforming back to configuration space, and writing $g_\mu \equiv G^{\sigma}{}_{\mu\sigma}$ for the trace of G and

$\mathrm{tr}(G_\mu G_\nu) \equiv G^\lambda{}_{\mu\sigma} G^\sigma{}_{\nu\lambda}$ for the matrix trace over the upper and one lower index, the action selected by the identities reads

$$S = \int d^d x \Big[- G^\lambda{}_{\mu\nu} \partial_\lambda \phi^{\mu\nu} + \big(\eta + \kappa\,\phi\big)^{\mu\nu} \Big(- \mathrm{tr}(G_\mu G_\nu) + \frac{1}{d-1}\, g_\mu g_\nu \Big) \Big], \tag{6.1}$$

which gives the Goldberg–Palatini action $\frac{1}{\kappa^2} \int d^d x\, h^{\mu\nu} R_{\mu\nu}(\Gamma)$, up to a surface term, written in the variables (2.1), with $h^{\mu\nu} = \eta^{\mu\nu} + \kappa \phi^{\mu\nu}$; Ref. [10] writes the same action with $\kappa = 1$, Eqs. (3.5)–(3.6) there. With the field dimensions and coupling-power assignment adopted here, the overall constant left undetermined by the identities is dimensionless. The explicit momentum-space kernels are given in Appendix E.

VII. REDUCTION TO THE METRIC FORMULATION

Writing the GG term of Eq. (6.1) as $\frac{1}{2} M(\eta + \kappa\phi)(G, G)$, with $M(h)(G, G) = 2h^{\mu\nu}\big(- \mathrm{tr}(G_\mu G_\nu) + \frac{1}{d-1} g_\mu g_\nu \big)$ linear in h, the field equation of G is algebraic, $M(\eta + \kappa\phi)\, G = \partial\phi$. For $d > 2$ and nondegenerate h its solution, substituted back, gives

$$S = -\tfrac{1}{2} \int d^d x\; M^{-1}(\eta + \kappa\phi)\big(\partial\phi, \partial\phi\big), \tag{7.1}$$

the second-order Einstein–Hilbert action in the Goldberg variable to all orders in ϕ, up to a surface term, Eq. (3.13) of Ref. [10]. The formal expansion $M^{-1}(\eta + \kappa\phi) = M^{-1}(\eta) - \kappa\, M^{-1}(\eta) M(\phi) M^{-1}(\eta) + \ldots$, Eq. (3.18) there, generates the quadratic action and every n-point vertex; with $G_0 \equiv M^{-1}(\eta)\, \partial\phi$, the cubic one is

$$S^{(3)} = \frac{\kappa}{2} \int d^d x\; M(\phi)\big(G_0, G_0\big). \tag{7.2}$$

The reduction has been checked symbolically at quadratic and cubic order against the Lagrangians of Capper, Leibbrandt and Ramón Medrano (CLM) [16], Eqs. (2.10) and (2.16) of that reference, which are written in the same metric-density perturbation $\sqrt{-g}\, g^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$. Starting from the bootstrap kernels, the accompanying notebook, written with xAct [17–19] (Appendix F), derives M_0^{-1}, with $M_0 \equiv M(\eta)$, verifies the left and right inverse relations, constructs $G_0 = M_0^{-1}\partial\phi$, and evaluates the quadratic Lagrangian from Eq. (7.1) and the cubic one from Eq. (7.2). The conventions must be kept explicit: CLM write $2K^{-2} \int d^d x\, h^{\mu\nu} R^{\mathrm{CLM}}_{\mu\nu}$ with $R^{\mathrm{CLM}}_{\mu\nu} = -R_{\mu\nu}$ in our convention (Eqs. (2.1) and (2.2) of Ref. [16]), whereas Eq. (6.1) corresponds to $\kappa^{-2} \int d^d x\, h^{\mu\nu} R_{\mu\nu}$ up to a surface term. With

$K = \kappa$ and $\mathcal{L}^{\rm CLM}_{(j)}$ the coefficients of their expansion $\mathcal{L} = \sum_{j\geq 2} K^{j-2}\mathcal{L}^{\rm CLM}_{(j)}$, the symbolic checks give

$$\mathcal{L}^{(2)} + \tfrac{1}{2}\,\mathcal{L}^{\rm CLM}_{(2)} = \partial_\mu J^\mu, \qquad J^\mu = \tfrac{1}{2}\big(\phi^{\nu\rho}\partial_\rho\phi^\mu{}_\nu - \phi^{\mu\nu}\partial_\rho\phi^\rho{}_\nu\big), \tag{7.3a}$$

$$\mathcal{L}^{(3)} + \tfrac{\kappa}{2}\,\mathcal{L}^{\rm CLM}_{(3)} = 0, \tag{7.3b}$$

with indices raised and lowered with η.[3] The cubic equality is local and requires neither integration by parts nor the field equations of ϕ; the quadratic actions agree under boundary conditions that remove the integral of $\partial_\mu J^\mu$. The surface term reflects the form in which CLM write their quadratic Lagrangian: expressed in the full density $h^{\mu\nu}$, the action obtained by eliminating G from Eq. (6.1) coincides identically with Eq. (A14) of Ref. [16], after accounting for the relative normalization stated above and without integration by parts, so that their Eq. (2.10) differs from the expansion of their Eq. (A14) by a total derivative. The corresponding three-graviton vertices therefore agree once the action, Fourier and symmetrization conventions are matched, without a separate transcription of CLM's momentum-space vertex.

Range of validity. Uniqueness holds within the class defined by the inputs listed in Section I; it is not established in a class containing further powers of the fields, additional derivatives, or independent dimensionful couplings. The overall constant is not fixed by the homogeneous invariance conditions; it requires an external normalization, such as matching the weak-field equations to the Newtonian limit with a specified normalization of the matter source. The construction is classical throughout; whether the first- and second-order formulations remain equivalent beyond tree level, a question studied in Ref. [10], is not addressed here.

VIII. CONCLUSIONS

The procedure described here identifies the Goldberg–Palatini action through the Noether identities of its prescribed field transformations. Within the local, Lorentz-covariant, parity-even ansatz, with the coupling-power assignment that bounds the derivative order and the cubic sector restricted to ϕGG, the 31 coefficients of the kernel bases (for $d \geq 4$) are reduced

[3] In transcribing Eq. (2.16) of Ref. [16], the first term of its second group is read as $\phi_{\mu\tau}\phi_{\mu\sigma,\lambda}\phi_{\lambda\tau,\sigma}$, which restores consistent index contractions in the printed expression.

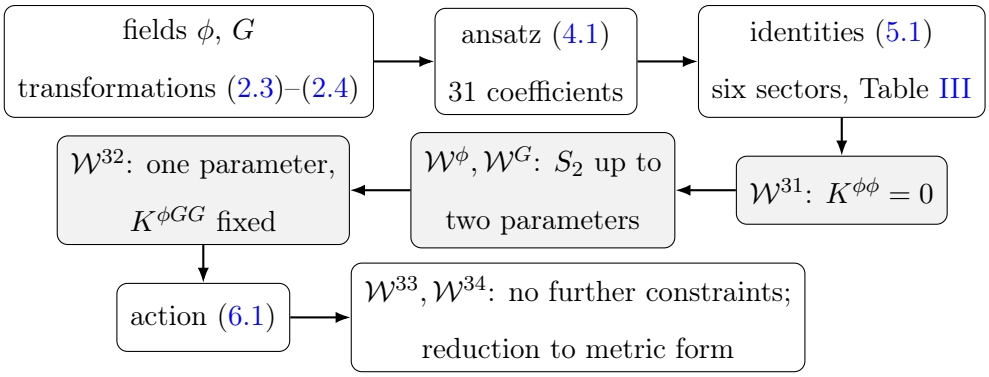


FIG. 1. The determination of the action. Inputs on the top row; the sequential solution of Table IV on the middle row; output and checks on the bottom row.

to a single overall constant. In particular, the identities eliminate the quadratic $\phi\phi$ kernel, which within this ansatz would only duplicate the graviton kinetic term generated by the elimination of G, and fix the relative coefficients of the mixed ϕG term, the algebraic GG term, and the cubic ϕGG interaction. The resulting structure is that of first-order Einstein gravity.

Individual Noether identities constrain only tensor contractions and can leave components of a kernel undetermined. Uniqueness here follows from imposing all identities simultaneously: the field-dependent transformations link the quadratic and cubic sectors and remove the freedom left by the quadratic identities alone, and the resulting homogeneous linear system has a one-dimensional solution space. Uniqueness is thus a property of this restricted system, not a general consequence of Noether identities. The connection is not a field exempt from the procedure: its variation enters every sector of the identity except the $\phi\phi$ one, and once the preceding constraints are imposed the cubic vertex is fixed by the sector that couples the two fields, $\mathcal{W}^{32}$.

Since the equation for G is algebraic and uniquely solvable for $d > 2$ and nondegenerate metric density, a construction that closes at cubic order in the first-order variables determines, through Eq. (7.1), the infinite set of vertices of the metric formulation; once the conventions are matched as in Section VII, its quadratic and cubic terms agree with those of Ref. [16], the quadratic ones up to a surface term. This is the practical content of the classical equivalence between the two formulations, and it is what makes the first-order variables the natural setting for the determination.

The method makes the role of symmetry explicit at each order in the fields. It avoids selecting an energy–momentum tensor as the source of successive self-couplings and instead imposes off-shell relations directly on the action kernels. Its practical advantage is that the polynomial first-order formulation turns the determination of the action into a finite tensor-algebra problem.

A natural extension is to enlarge the interaction ansatz while retaining explicit control over locality and derivative order. The first step is to admit the cubic structures $\phi\phi G$ and $\phi\phi\phi$ allowed by the derivative counting and to determine whether the identities set them to zero or require terms of higher order in the fields; this would test whether the restriction to ϕGG can be relaxed without introducing new invariant structures. For Yang–Mills theory the tree-amplitude version of the argument is known to close and to fix the interaction structure under its stated assumptions [8]; the action-level version with tensor bases would provide a direct check of the method, while higher-derivative gravitational actions would allow it to probe a broader space of gauge-invariant dynamics.

ACKNOWLEDGMENTS

F.T.B. acknowledges partial support from CNPq (grant No. 309411/2022-0). A.C.B.C. acknowledges an undergraduate research scholarship from the Programa Unificado de Bolsas de Estudos of the University of São Paulo (PUB-USP). In preparing this manuscript the authors used AI tools as assistants for literature search, language and LaTeX editing, and coding.

Appendix A: Notation and transformation laws

Table V lists the objects used in the main text, with their index structure, momentum dependence and mass dimension.

The transformation laws (2.3) and (2.4) are Eqs. (17b) and (18) of Ref. [13], where $\kappa = 1$ and the parameter θ is $-\kappa\omega$ here. We sketch their derivation. Under $x'^{\mu} = x^{\mu} - \varepsilon^{\mu}(x)$ the variation $\delta T(x) \equiv T'(x) - T(x)$ of a tensor is its Lie derivative $\mathcal{L}_{\varepsilon} T$; for a $(1,2)$ tensor,

$$\mathcal{L}_{\varepsilon} X^{\lambda}{}_{\mu\nu} = \varepsilon^{\rho}\partial_{\rho} X^{\lambda}{}_{\mu\nu} - X^{\rho}{}_{\mu\nu}\partial_{\rho}\varepsilon^{\lambda} + X^{\lambda}{}_{\rho\nu}\partial_{\mu}\varepsilon^{\rho} + X^{\lambda}{}_{\mu\rho}\partial_{\nu}\varepsilon^{\rho}, \tag{A1}$$

and $\mathcal{L}_{\varepsilon}$ annihilates the Kronecker delta.

Symbol	Meaning	Indices	Momenta	Dimension
κ	gravitational coupling			$(2-d)/2$
$\phi^{\mu\nu}$; ϕ_p	density perturbation, Eq. (2.1)	$\mu\nu$ symmetric	one	$(d-2)/2$; $-(d+2)/2$
$G^{\lambda}{}_{\mu\nu}$; G_p	trace-adjusted connection, Eq. (2.1)	$\mu\nu$ symmetric	one	$d/2$; $-d/2$
ω^{μ}; ω_p	gauge parameter	vector	one	$(d-4)/2$; $-(d+4)/2$
$D_{\phi^0}(p)$, $D_{G^0}(p)$	$\delta^{(0)}\phi$, $\delta^{(0)}G$ acting on ω	App. B	p	1, 2
$D_{\phi^1}(p,q)$, $D_{G^1}(p,q)$	$\delta^{(1)}\phi$, $\delta^{(1)}G$ acting on (field, ω)	App. B	p,q	1, 1
$K^{\phi\phi}$, $K^{\phi G}$, K^{GG}	quadratic kernels of S_2	Table II	p,p, none	2, 1, 0
$K^{\phi GG}$	cubic kernel of S_3	Table II	none	0
$\mathcal{W}^{\phi}, \mathcal{W}^{G}, \mathcal{W}^{31}, \ldots, \mathcal{W}^{34}$	the six Noether identities	App. D	p; p_i	

TABLE V. Notation and mass dimensions. For the fields and the gauge parameter the two entries in the last column refer to configuration and momentum space; $[\int_p] = d$.

a. The metric density. $h^{\mu\nu} \equiv \sqrt{-g}\, g^{\mu\nu}$ is a contravariant tensor density of weight one,

$$\delta h^{\mu\nu} = \partial_\rho\big(h^{\mu\nu}\varepsilon^\rho\big) - h^{\rho\nu}\partial_\rho\varepsilon^\mu - h^{\mu\rho}\partial_\rho\varepsilon^\nu, \tag{A2}$$

which is Eq. (17a) of Ref. [13]. Holding the background $\eta^{\mu\nu}$ fixed, so that $\delta h^{\mu\nu} = \kappa\,\delta\phi^{\mu\nu}$, inserting $h^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$ and $\varepsilon = \kappa\omega$, and dividing by κ gives Eq. (2.3).

b. The connection. An affine connection transforms as

$$\delta\Gamma^{\lambda}{}_{\mu\nu} = \mathcal{L}_\varepsilon\Gamma^{\lambda}{}_{\mu\nu} + \partial_\mu\partial_\nu\varepsilon^\lambda, \tag{A3}$$

where, here and below, $\mathcal{L}_\varepsilon$ applied to a non-tensorial object denotes only the tensor-form expression (A1). Contracting λ with μ, the trace $T_\nu \equiv \Gamma^{\lambda}{}_{\nu\lambda}$ obeys

$$\delta T_\nu = \mathcal{L}_\varepsilon T_\nu + \partial_\nu(\partial\cdot\varepsilon). \tag{A4}$$

By Eq. (2.1), $\delta(\kappa G^{\lambda}{}_{\mu\nu}) = \delta\Gamma^{\lambda}{}_{\mu\nu} - \frac{1}{2}\big(\delta^\lambda_\mu\,\delta T_\nu + \delta^\lambda_\nu\,\delta T_\mu\big)$. Since $\mathcal{L}_\varepsilon$ annihilates δ^λ_μ, the tensor-form terms combine into $\kappa\,\mathcal{L}_\varepsilon G$, and only the inhomogeneous part of Eq. (A4) survives from the trace correction:

$$\delta(\kappa G^{\lambda}{}_{\mu\nu}) = \kappa\,\mathcal{L}_\varepsilon G^{\lambda}{}_{\mu\nu} + \partial_\mu\partial_\nu\varepsilon^\lambda - \tfrac{1}{2}\big(\delta^\lambda_\mu\partial_\nu + \delta^\lambda_\nu\partial_\mu\big)\partial\cdot\varepsilon, \tag{A5}$$

which with $\varepsilon = \kappa\omega$ is Eq. (2.4). Its trace, $\delta^{(0)}G^{\lambda}{}_{\nu\lambda} = \frac{1-d}{2}\,\partial_\nu\partial\cdot\omega$, is consistent with $\Gamma^{\sigma}{}_{\nu\sigma} = \frac{2\kappa}{1-d}g_\nu$. Both Eq. (A4) and the final result were verified symbolically (Appendix F).

Appendix B: Transformations in momentum space

With the Fourier conventions of Section II, Eq. (2.5) reads, with all indices displayed,

$$\delta\tilde{\phi}^{\mu\nu}(p) = D_{\phi^0}{}^{\mu\nu}{}_{\alpha}(p)\,\tilde{\omega}^{\alpha}(p) + \kappa \int_q \tilde{\phi}^{\alpha\beta}(p-q)\,\tilde{\omega}^{\rho}(q)\,D_{\phi^1}{}^{\mu\nu}{}_{\alpha\beta\rho}(p,q), \tag{B1}$$

$$\delta\tilde{G}^{\lambda}{}_{\mu\nu}(p) = D_{G^0}{}^{\lambda}{}_{\mu\nu\alpha}(p)\,\tilde{\omega}^{\alpha}(p) + \kappa \int_q \tilde{G}^{\alpha}{}_{\beta\gamma}(p-q)\,\tilde{\omega}^{\rho}(q)\,D_{G^1}{}^{\lambda}{}_{\mu\nu\alpha}{}^{\beta\gamma}{}_{\rho}(p,q), \tag{B2}$$

with

$$D_{\phi^0}{}^{\mu\nu}{}_{\alpha}(p) = -i\big(\eta^{\mu\nu}p_\alpha - p^{(\mu}\delta^{\nu)}_{\alpha}\big), \tag{B3}$$

$$D_{\phi^1}{}^{\mu\nu}{}_{\alpha\beta\rho}(p,q) = -i\big(\delta^{\mu}_{\alpha}\delta^{\nu}_{\beta}\,p_\rho - q_\beta\,\delta^{(\mu}_{\alpha}\delta^{\nu)}_{\rho}\big), \tag{B4}$$

$$D_{G^0}{}^{\lambda}{}_{\mu\nu\alpha}(p) = -p_\mu p_\nu \delta^{\lambda}_{\alpha} + \tfrac{1}{2}\delta^{\lambda}_{(\mu}p_{\nu)}p_\alpha, \tag{B5}$$

$$D_{G^1}{}^{\lambda}{}_{\mu\nu\alpha}{}^{\beta\gamma}{}_{\rho}(p,q) = -i\big[\delta^{\lambda}_{\alpha}\delta^{\beta}_{\mu}\delta^{\gamma}_{\nu}\,(p-q)_\rho - q_\alpha\,\delta^{\beta}_{\mu}\delta^{\gamma}_{\nu}\delta^{\lambda}_{\rho} + \delta^{\lambda}_{\alpha}\delta^{\beta}_{\rho}\delta^{\gamma}_{(\mu}q_{\nu)}\big]. \tag{B6}$$

These expressions are understood as acting on fields with their stated index symmetries; symmetrization in the contracted field indices is therefore implicit, and a representative differing by terms that vanish on symmetric fields gives the same result.

Appendix C: Tensor bases

Each kernel is expanded in the basis of all independent tensors with its index structure, built from η and p with the momentum dependence of Table II. The construction is the same for the four kernels. One enumerates the tensor monomials of the required degree in p, each a single product of η's over the index positions of K not carrying a momentum, times the appropriate powers of p, and symmetrizes each monomial under the index symmetries of the kernel. These symmetries, listed in Table II, generate a finite group G_K of permutations of the index positions of K. Applying every element of G_K to a monomial T produces a set of n distinct monomials $T_1, \ldots, T_n$, the *orbit* of T; the symmetrized monomial is the average over the orbit, $\mathcal{S}[T] = \frac{1}{n}\sum_{k=1}^{n} T_k$ (equivalently, the average of T over the group). Two monomials give the same $\mathcal{S}[T]$ if and only if they lie in the same orbit, so the orbit averages form a basis of the invariant polynomials in η and p, in one-to-one correspondence with the orbits, and any element of an orbit may serve as its representative. This is a candidate tensor basis; its independence as a set of tensors in a given dimension is checked below. The orbit sizes n add up to the number of monomials. In the implementation the monomials are generated

by applying all permutations of the index positions to a few prototensors (the distinct ways of distributing η's and momenta) and removing duplicates; the symmetrized elements are then obtained by averaging over the kernel symmetries and removing duplicates again, which yields exactly the orbit averages described above.

Tables VI–IX list, for each kernel, a representative monomial of every orbit and the number n of terms in the corresponding symmetrized sum. Indices are written in the order of the kernel's index positions; $p^2 \equiv p_\alpha p^\alpha$. For $K^{\phi\phi}$, for instance, element 4 of Table VI is $\frac{1}{4}\big(p_\mu p_\rho \eta_{\nu\sigma} + p_\nu p_\rho \eta_{\mu\sigma} + p_\mu p_\sigma \eta_{\nu\rho} + p_\nu p_\sigma \eta_{\mu\rho}\big)$.

Distinct orbits have disjoint monomial content, so the elements are linearly independent as polynomials in η and p, before any dimension-dependent identity is used. They can nevertheless become dependent as tensors in low dimension, through identities that hold only for specific d. Independence for generic d is verified by computing the Gram matrix Γ_K of each basis, $(\Gamma_K)_{ij} = \mathcal{S}[T_i] \cdot \mathcal{S}[T_j]$ with all indices contracted. Its determinant is a polynomial in d and p^2: when it is nonzero the basis is linearly independent, in any signature, and its zeros locate the dimensions where the basis degenerates (at Euclidean momentum the contraction pairing is positive definite, so there a vanishing determinant implies a dependence). The bases used here depend on a single momentum and their elements share the same degree in p, so a dependence with coefficients depending on p^2 is equivalent to one with constant coefficients; with several independent momenta, a vanishing Gram determinant need not imply a dependence with constant coefficients. For the bases of Tables VI–IX, up to positive numerical factors,

$$\det \Gamma_{\phi\phi} = (p^2)^8\,(d-2)(d-1)^3(d+2)(d+4), \tag{C1}$$

$$\det \Gamma_{\phi G} = (p^2)^6\,(d-2)(d-1)^5(d+2)^5(d+4), \tag{C2}$$

$$\det \Gamma_{GG} = d^5\,(d-2)(d-1)^4(d+2)^4(d+4), \tag{C3}$$

$$\det \Gamma_{\phi GG} = d^{16}\,(d-3)(d-2)^7(d-1)^{15}(d+1)^3(d+2)^{15}(d+4)^6(d+6). \tag{C4}$$

All four are nonzero for $d \geq 4$ and $p^2 \neq 0$; the powers of p^2 reflect the momentum dependence of the first two bases and are unrelated to dimension-dependent relations. The zeros at $d = 1, 2$, and at $d = 3$ for $K^{\phi GG}$, are the low dimensions in which products of η's with many indices become linearly dependent (in $d = 3$, for instance, antisymmetrization over four indices vanishes).

	Representative monomial	n
1	$p^2\,\eta_{\mu\nu}\eta_{\rho\sigma}$	1
2	$p^2\,\eta_{\mu\rho}\eta_{\nu\sigma}$	2
3	$p_\mu p_\nu \eta_{\rho\sigma}$	2
4	$p_\mu p_\rho \eta_{\nu\sigma}$	4

TABLE VI. Basis of $K^{\phi\phi}_{\mu\nu\rho\sigma}(p)$: 4 elements, the orbits of the 9 monomials under the symmetry group of order 8.

	Representative monomial	n
1	$p_\mu \eta_{\nu\lambda}\eta_{\rho\sigma}$	2
2	$p_\mu \eta_{\nu\rho}\eta_{\lambda\sigma}$	4
3	$p_\lambda \eta_{\mu\nu}\eta_{\rho\sigma}$	1
4	$p_\lambda \eta_{\mu\rho}\eta_{\nu\sigma}$	2
5	$p_\rho \eta_{\mu\nu}\eta_{\lambda\sigma}$	2
6	$p_\rho \eta_{\mu\lambda}\eta_{\nu\sigma}$	4

TABLE VII. Basis of $K^{\phi G}_{\mu\nu\lambda\rho\sigma}(p)$: 6 elements, the orbits of the 15 monomials under the symmetry group of order 4.

Appendix D: The six identities

In the notation of Section II, each identity of Table III states that an expression linear in each field and in the gauge parameter vanishes identically. Writing P for the total momentum

	Representative monomial	n
1	$\eta_{\lambda\mu}\eta_{\nu\rho}\eta_{\sigma\tau}$	4
2	$\eta_{\lambda\mu}\eta_{\nu\sigma}\eta_{\rho\tau}$	4
3	$\eta_{\lambda\rho}\eta_{\mu\nu}\eta_{\sigma\tau}$	1
4	$\eta_{\lambda\rho}\eta_{\mu\sigma}\eta_{\nu\tau}$	2
5	$\eta_{\lambda\sigma}\eta_{\mu\rho}\eta_{\nu\tau}$	4

TABLE VIII. Basis of $K^{GG}_{\lambda\mu\nu\rho\sigma\tau}$: 5 elements, the orbits of the 15 monomials under the symmetry group of order 8.

	Representative monomial	n		Representative monomial	n
1	$\eta_{\mu\nu}\eta_{\lambda\alpha}\eta_{\beta\rho}\eta_{\sigma\tau}$	4	9	$\eta_{\mu\lambda}\eta_{\nu\rho}\eta_{\alpha\sigma}\eta_{\beta\tau}$	4
2	$\eta_{\mu\nu}\eta_{\lambda\alpha}\eta_{\beta\sigma}\eta_{\rho\tau}$	4	10	$\eta_{\mu\lambda}\eta_{\nu\sigma}\eta_{\alpha\beta}\eta_{\rho\tau}$	8
3	$\eta_{\mu\nu}\eta_{\lambda\rho}\eta_{\alpha\beta}\eta_{\sigma\tau}$	1	11	$\eta_{\mu\lambda}\eta_{\nu\sigma}\eta_{\alpha\rho}\eta_{\beta\tau}$	16
4	$\eta_{\mu\nu}\eta_{\lambda\rho}\eta_{\alpha\sigma}\eta_{\beta\tau}$	2	12	$\eta_{\mu\alpha}\eta_{\nu\beta}\eta_{\lambda\rho}\eta_{\sigma\tau}$	4
5	$\eta_{\mu\nu}\eta_{\lambda\sigma}\eta_{\alpha\rho}\eta_{\beta\tau}$	4	13	$\eta_{\mu\alpha}\eta_{\nu\beta}\eta_{\lambda\sigma}\eta_{\rho\tau}$	8
6	$\eta_{\mu\lambda}\eta_{\nu\alpha}\eta_{\beta\rho}\eta_{\sigma\tau}$	8	14	$\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\beta}\eta_{\rho\tau}$	8
7	$\eta_{\mu\lambda}\eta_{\nu\alpha}\eta_{\beta\sigma}\eta_{\rho\tau}$	16	15	$\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\rho}\eta_{\beta\tau}$	8
8	$\eta_{\mu\lambda}\eta_{\nu\rho}\eta_{\alpha\beta}\eta_{\sigma\tau}$	2	16	$\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\tau}\eta_{\beta\rho}$	8

TABLE IX. Basis of $K^{\phi GG}_{\mu\nu\lambda\alpha\beta\rho\sigma\tau}$: 16 elements, the orbits of the 105 monomials (one for each way of pairing the eight indices into four η's) under the symmetry group of order 16.

($P = p_1 + p_2$ or $p_1 + p_2 + p_3$), the six forms are

$$\mathcal{W}^{\phi}: \quad 2K^{\phi\phi}\big(\phi_{-p}, D_{\phi^0}(p)\,\omega_p\big) + K^{\phi G}\big(\phi_{-p}, D_{G^0}(p)\,\omega_p\big) = 0, \tag{D1}$$

$$\mathcal{W}^{G}: \quad K^{\phi G}\big(D_{\phi^0}(p)\,\omega_p, G_{-p}\big) + 2K^{GG}\big(G_{-p}, D_{G^0}(p)\,\omega_p\big) = 0, \tag{D2}$$

$$\mathcal{W}^{31}: \quad K^{\phi\phi}\big(\phi_{-p_1}, D_{\phi^1}(p_1, P)(\phi_{-p_2}, \omega_P)\big) + (1 \leftrightarrow 2) = 0, \tag{D3}$$

$$\begin{aligned}\mathcal{W}^{32}: \quad & 2K^{\phi GG}\big(\phi_{-p_2}, D_{G^0}(P)\,\omega_P, G_{-p_1}\big) + K^{\phi G}\big(D_{\phi^1}(p_1, P)(\phi_{-p_2}, \omega_P), G_{-p_1}\big) \\ & + K^{\phi G}\big(\phi_{-p_2}, D_{G^1}(p_2, P)(G_{-p_1}, \omega_P)\big) = 0,\end{aligned} \tag{D4}$$

$$\mathcal{W}^{33}: \quad K^{\phi GG}\big(D_{\phi^0}(P)\,\omega_P, G_{-p_1}, G_{-p_2}\big) + K^{GG}\big(G_{-p_1}, D_{G^1}(p_1, P)(G_{-p_2}, \omega_P)\big) + (1 \leftrightarrow 2) = 0, \tag{D5}$$

$$\begin{aligned}\mathcal{W}^{34}: \quad & K^{\phi GG}\big(\phi_{-p_3}, G_{-p_1}, D_{G^1}(p_1 + p_3, P)(G_{-p_2}, \omega_P)\big) + (1 \leftrightarrow 2) \\ & + K^{\phi GG}\big(D_{\phi^1}(p_1 + p_2, P)(\phi_{-p_3}, \omega_P), G_{-p_1}, G_{-p_2}\big) = 0,\end{aligned} \tag{D6}$$

where $(1 \leftrightarrow 2)$ adds the immediately preceding term with the two field arguments of the same type exchanged, that is, with $p_1 \leftrightarrow p_2$ and, in the component form below, with their index groups exchanged as well; in $\mathcal{W}^{33}$ it applies only to the K^{GG} term. The factors of two count the two equal contributions of a kernel that is quadratic in the field being varied. Each term is one kernel of Eq. (4.1) with one argument replaced by the appropriate piece of the transformation (2.5), the momenta being fixed by momentum conservation. Since the fields are arbitrary within their symmetry classes, the coefficient tensor of $\phi_{-p_1} \cdots \omega_P$ in each form, projected onto the index symmetries of the field indices, vanishes. The scalar forms above only define the identities and are not used as such: we work throughout with these coefficient tensors, whose free indices are kept so that each identity can be projected onto the bases described below. With all indices displayed, and in terms of the tensors of

Appendix B, these coefficient tensors are

$$\mathcal{W}^{\phi}_{\rho\sigma\alpha}(p) \equiv 2D_{\phi^0}{}^{\mu\nu}{}_{\alpha}(p)K^{\phi\phi}_{\mu\nu\rho\sigma}(p) + D_{G^0}{}^{\lambda}{}_{\mu\nu\alpha}(p)K^{\phi G\,\mu\nu}_{\rho\sigma\lambda}(p) = 0, \tag{D7}$$

$$\mathcal{W}^{G}_{\rho\sigma\tau\alpha}(p) \equiv D_{\phi^0}{}^{\mu\nu}{}_{\alpha}(p)K^{\phi G}_{\mu\nu\rho\sigma\tau}(-p) + 2D_{G^0}{}^{\lambda}{}_{\mu\nu\alpha}(p)K^{GG\mu\nu}_{\lambda\ \rho\sigma\tau} = 0, \tag{D8}$$

$$\begin{aligned}\mathcal{W}^{31}_{\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & K^{\phi\phi}_{\mu_1\nu_1\alpha\beta}(p_1)D_{\phi^1}{}^{\alpha\beta}{}_{\mu_2\nu_2\rho}(p_1,p_1+p_2) \\ & + K^{\phi\phi}_{\mu_2\nu_2\alpha\beta}(p_2)D_{\phi^1}{}^{\alpha\beta}{}_{\mu_1\nu_1\rho}(p_2,p_1+p_2) = 0,\end{aligned} \tag{D9}$$

$$\begin{aligned}\mathcal{W}^{32}_{\lambda_1\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & 2K^{\phi GG}_{\mu_2\nu_2\lambda\mu\nu\lambda_1\mu_1\nu_1}D_{G^0}{}^{\lambda\mu\nu}{}_{\rho}(p_1+p_2) \\ & + K^{\phi G}_{\mu\nu\lambda_1\mu_1\nu_1}(-p_1)D_{\phi^1}{}^{\mu\nu}{}_{\mu_2\nu_2\rho}(p_1,p_1+p_2) \\ & + K^{\phi G}_{\mu_2\nu_2\lambda}{}^{\mu\nu}(p_2)D_{G^1}{}^{\lambda}{}_{\mu\nu\lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2) = 0,\end{aligned} \tag{D10}$$

$$\begin{aligned}\mathcal{W}^{33}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}D_{\phi^0}{}^{\mu\nu}{}_{\rho}(p_1+p_2) \\ & + K^{GG}_{\lambda\mu\nu\lambda_1\mu_1\nu_1}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1,p_1+p_2) \\ & + K^{GG}_{\lambda\mu\nu\lambda_2\mu_2\nu_2}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2) = 0,\end{aligned} \tag{D11}$$

$$\begin{aligned}\mathcal{W}^{34}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\mu_3\nu_3\rho}(p_1,p_2,p_3) \equiv & \\ & K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_1\mu_1\nu_1}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1+p_3,P) \\ & + K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_2\mu_2\nu_2}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2+p_3,P) \\ & + K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}D_{\phi^1}{}^{\mu\nu}{}_{\mu_3\nu_3\rho}(p_1+p_2,P) = 0,\end{aligned} \tag{D12}$$

The identities above retain their generic momentum dependence. For the projection bases of the cubic identities the independent momenta are identified, $p_1 = p_2 = p$, and also $p_3 = p$ in $\mathcal{W}^{34}$, as explained in Section V. The projection bases of Table III are generated by the procedure of Appendix C, from the monomials with the index structure and degree in p of each identity, averaged over its symmetry group: the symmetries of the kernel indices involved and, for $\mathcal{W}^{31}$, $\mathcal{W}^{33}$ and $\mathcal{W}^{34}$, the exchange of the two identical fields. Table X lists the three smallest ones as an illustration; the remaining dimensions, 23, 20 and 95, make an explicit listing unhelpful, and those bases are generated by the code of Appendix F. Their Gram determinants are, up to numerical factors, $(p^2)^9(d-1)^2$, $(p^2)^{12}(d-2)(d-1)^5(d+2)(d+4)$ and $(p^2)^{21}(d-2)^2(d-1)^6(d+1)(d+4)(d+6)$ for the three bases of Table X, and for $p^2 \neq 0$ those of the bases of $\mathcal{W}^{32}$ and $\mathcal{W}^{33}$ vanish, among positive integer dimensions, only at $d = 1, 2, 3$; all five are therefore nondegenerate for $d \geq 4$ and $p^2 \neq 0$. The basis of $\mathcal{W}^{34}$ becomes degenerate at $d = 4$. This does not obstruct the determination, since no inversion of that Gram matrix is needed: the other identities fix the kernels, $\mathcal{W}^{34}$ is then found to be

satisfied identically by substitution, and its vanishing also follows from the identification of the solution with the invariant Goldberg–Palatini action.

Representative monomial	n	Representative monomial	n
$\mathcal{W}^{\phi}_{\rho\sigma\alpha}(p)$		$\mathcal{W}^{31}_{\mu_1\nu_1\mu_2\nu_2\rho}(p,p)$	
1 $p_\rho p_\sigma p_\alpha$	1	1 $p_{\mu_1} p_{\nu_1} p_\rho \eta_{\mu_2\nu_2}$	2
2 $p^2 p_\alpha \eta_{\rho\sigma}$	1	2 $p_{\mu_1} p_{\nu_1} p_{\mu_2} \eta_{\nu_2\rho}$	4
3 $p^2 p_\rho \eta_{\sigma\alpha}$	2	3 $p_{\mu_1} p_{\mu_2} p_\rho \eta_{\nu_1\nu_2}$	4
$\mathcal{W}^{G}_{\rho\sigma\tau\alpha}(p)$		4 $p^2 p_\rho \eta_{\mu_1\nu_1} \eta_{\mu_2\nu_2}$	1
1 $p_\rho p_\alpha \eta_{\sigma\tau}$	1	5 $p^2 p_\rho \eta_{\mu_1\mu_2} \eta_{\nu_1\nu_2}$	2
2 $p_\sigma p_\tau \eta_{\rho\alpha}$	1	6 $p^2 p_{\mu_1} \eta_{\nu_1\rho} \eta_{\mu_2\nu_2}$	4
3 $p_\rho p_\sigma \eta_{\tau\alpha}$	2	7 $p^2 p_{\mu_1} \eta_{\nu_1\mu_2} \eta_{\nu_2\rho}$	8
4 $p_\sigma p_\alpha \eta_{\rho\tau}$	2		
5 $p^2 \eta_{\rho\alpha} \eta_{\sigma\tau}$	1		
6 $p^2 \eta_{\rho\sigma} \eta_{\tau\alpha}$	2		

TABLE X. The three smallest projection bases, in the notation of Appendix C: 3, 6 and 7 elements, the orbits of 4, 9 and 25 monomials under symmetry groups of order 2, 2 and 8. For $\mathcal{W}^{31}$ the two momenta are identified, $p_1 = p_2 = p$, and the symmetry group includes the exchange $(\mu_1\nu_1) \leftrightarrow (\mu_2\nu_2)$.

Appendix E: Explicit kernels

Up to the overall constant, the solution of Table IV is, in terms of the basis elements $\mathcal{S}[T]$ of Appendix C (the average of the monomial T over its orbit), and with all indices of

the kernels lowered, to be raised with $\eta^{\mu\nu}$ as required by the contractions with the fields,

$$K^{\phi\phi}_{\mu\nu\rho\sigma}(p) = 0, \tag{E1}$$

$$K^{\phi G}_{\mu\nu\lambda\rho\sigma}(p) = -i\,\mathcal{S}\big[p_\lambda \eta_{\mu\rho}\eta_{\nu\sigma}\big] = -\tfrac{i}{2}\,p_\lambda\big(\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}\big), \tag{E2}$$

$$\begin{aligned} K^{GG}_{\lambda\mu\nu\rho\sigma\tau} &= -\mathcal{S}\big[\eta_{\lambda\sigma}\eta_{\mu\rho}\eta_{\nu\tau}\big] + \frac{1}{d-1}\,\mathcal{S}\big[\eta_{\lambda\mu}\eta_{\nu\sigma}\eta_{\rho\tau}\big] \\ &= \tfrac{1}{4}\Big[- \eta_{\lambda(\sigma}\eta_{\tau)(\mu}\eta_{\nu)\rho} + \frac{1}{d-1}\,\eta_{\lambda(\mu}\eta_{\nu)(\sigma}\eta_{\tau)\rho}\Big], \end{aligned} \tag{E3}$$

$$K^{\phi GG}_{\mu\nu\lambda\alpha\beta\rho\sigma\tau} = -\mathcal{S}\big[\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\tau}\eta_{\beta\rho}\big] + \frac{1}{d-1}\,\mathcal{S}\big[\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\beta}\eta_{\rho\tau}\big], \tag{E4}$$

that is, element 4 of Table VII, elements 5 and 2 of Table VIII, and elements 16 and 14 of Table IX; the latter two basis elements contain eight terms each. In the GG and ϕGG kernels the first structure contracts the connection index of one G with a lower index of the other, and the second is the product of the two traces $g_\mu \equiv G^\sigma{}_{\mu\sigma}$. Contracted with the fields, in the notation of Section II, these give

$$K^{\phi G}(\phi_{-p}, G_p) = -ip_\lambda\,\tilde\phi^{\mu\nu}(-p)\tilde G^\lambda{}_{\mu\nu}(p), \tag{E5}$$

$$K^{GG}(G_{-p}, G_p) = \eta^{\mu\nu}\Big(- \tilde G^\lambda{}_{\mu\sigma}(-p)\tilde G^\sigma{}_{\nu\lambda}(p) + \frac{1}{d-1}\tilde G^\sigma{}_{\mu\sigma}(-p)\tilde G^\rho{}_{\nu\rho}(p)\Big), \tag{E6}$$

$$K^{\phi GG}(\phi_{p+q}, G_{-p}, G_{-q}) = \tilde\phi^{\mu\nu}(p+q)\Big(- \tilde G^\lambda{}_{\mu\sigma}(-p)\tilde G^\sigma{}_{\nu\lambda}(-q) + \frac{1}{d-1}\tilde G^\sigma{}_{\mu\sigma}(-p)\tilde G^\rho{}_{\nu\rho}(-q)\Big), \tag{E7}$$

whose inverse Fourier transform yields Eq. (6.1).

Appendix F: Symbolic implementation and reproducibility

In the first-order bootstrap implementation the coupling is set to $\kappa = 1$. Since κ enters the construction only as the overall factor of S_3 in Eq. (4.1) and of $\delta^{(1)}$ in Eq. (2.5), this affects neither the kernels of Appendix E nor the identities of Appendix D; it is restored in the results by dimensional analysis, as in Section III. The second-order comparison keeps κ symbolic throughout.

a. Files. The calculations are organized in the following Mathematica [15] notebooks and data files, kept together in one directory.

1. `Goldberg_Palatini_Bootstrap_en.nb` (FeynCalc [14]): the first-order determination. It generates the tensor bases of the four kernels and of the six identities (Appendices C and D) and the Gram determinants of the kernel bases and of the five smallest

projection bases (the rank test of the 95-element basis of $\mathcal{W}^{34}$ at $d = 4$ is an optional diagnostic, not part of the main flow), builds the nonlinear identities with the field momenta identified, solves the linear system in the sequence of Table IV, checks the resulting kernels against Eqs. (3.5)–(3.6) of Ref. [10], verifies the complete cubic kernel against $\frac{1}{2}M(\phi)$ (cell tagged `FullCubicKernelCheck`), and exports the normalized kernels.

2. `Goldberg_Palatini_Bootstrap_kernels.wl`: the exported kernels K^{GG} and $K^{\phi GG}$ in a package-independent representation by index positions, with the conventions (dimension symbol, symmetries, normalization $c_{GG} = 1/(d-1)$, Fourier sign); it is data, not an executable script.

3. `Goldberg_Palatini_Cubic_CLM.nb` (xAct/xTensor, xTras and xPerm [17–19]): the second-order comparison of Section VII. It imports the data file without FeynCalc, converts it to xAct tensors, derives M_0^{-1} by solving its defining identity in a five-element basis of metric structures and checks both inverse relations (`inverseChecks`), substitutes G_0 back into $M_0 G_0 = \partial\phi$ as a test of index placement and normalization (`g0Check`), and evaluates the residuals of Eqs. (7.3) (`clmQuadraticCheck`, `quadraticBoundaryCheck`, `clmCubicResidual`); `allChecksPassed` only aggregates these residuals. A final section compares, in terms of the full density $h^{\mu\nu}$, the action obtained by eliminating G with Eq. (A14) of Ref. [16]. The notebook uses the metric-compatible derivative of the flat background, `CD`, which in Cartesian coordinates is the partial derivative of the text; the dimension is kept symbolic, and the pole of M_0^{-1} at $d = 2$ is separate from the range $d \geq 4$ of the bases.

A `README` gives the evaluation order (the first notebook and its export cell first; the comparison notebook in a fresh kernel), the required packages and versions, the expected result of every check, the conventions of the comparison, and the correspondence between the notebooks and the equations and tables of this paper.

The calculations were carried out with Mathematica 15.0.0 (macOS) and repeated with Mathematica 13.3.1 (macOS) and 14.2.0 (Linux), with identical results, using FeynCalc 10.1.0, xTensor 1.2.0, xPerm 1.2.3 and xTras 1.4.2; the notebooks require Mathematica and these packages, and other versions have not been systematically tested. The

notebooks, data file, README and the source of this paper are archived on Zenodo at https://doi.org/10.5281/zenodo.22757769.

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